

Intro to Linux

Author: @c3rkah & @MicahKBrown

Goal: Learn the fundamental of navigating and working with Linux.

Pre Requirements:

- Connect to the CTF wireless
- Personal laptop
- Preferable:
    - Download and install Virtual Box ahead of time.
    - Download and install / boot Linux of your choice

Linux Overview:

Over the past few decades we have seen a technological renaissance transform our world. One key but often silent enabler to this digital transformation has been the Open Source Operating name Linux. Over the years it has grown more powerful, flexible, and sustainable. Instead of product owners having to license their operating system, they could simply and freely leverage Linux. Today we find Linux in our servers, Cars, TVs, phones, digital displays, exercise machines, smart watches, and technology categories too many to list. Often vast swafts of a cloud infrastructure is built ontop of Linux. Often companies will hide the general Linux user interface and terminal (command line interface) behind a more user friendly user interface.

Lets learn a brief history of Linux from our friends at Wikipedia:

> “
>
> Linux (/ˈlɪnʊks/, LIN-uuks)[15] is a family of open-source Unix-like operating systems based on the Linux kernel,[16] an operating system kernel first released on September 17, 1991, by Linus Torvalds.[17][18][19] Linux is typically packaged as a Linux distribution (distro), which includes the kernel and supporting system software and libraries—most of which are provided by third parties—to create a complete operating system, designed as a clone of Unix and released under the copyleft GPL license.[20]
>
> Many Linux distributions use the word "Linux" in their name, but the Free Software Foundation uses and recommends the name "GNU/Linux" to emphasize the use and importance of GNU software in many

distributions, causing some controversy.[21][22] Thousands of distributions exist, many based directly or indirectly on other distributions;[23][24] popular Linux distributions[25][26][27] include Debian, Fedora Linux, Linux Mint, Arch Linux, and Ubuntu, while commercial distributions include Red Hat Enterprise Linux, SUSE Linux Enterprise, and ChromeOS. Linux distributions are frequently used in server platforms.[28][29] Other than the Linux kernel, key components that make up a distribution may include a display server (windowing system), a package manager, a bootloader and a Unix shell.

Linux is one of the most prominent examples of free and open-source software collaboration. While originally developed for x86 based personal computers, it has since been ported to more platforms than any other operating system,[30] and is used on a wide variety of devices including PCs, workstations, mainframes and embedded systems. Linux is the predominant operating system for servers and is also used on all of the world's 500 fastest supercomputers.[g] When combined with Android, which is Linux-based and designed for smartphones, they have the largest installed base of all general-purpose operating systems.
”

https://en.wikipedia.org/wiki/Linux

As Linux is open source, we have many popular distributions that we can choose from. It is important to remember that as each distribution is published independently, we will find differences in implementation, functionality, and performance. For the sake of this workshop we suggest that you start with one of these three popular Linux Distributions:

- Kali: Kali is a feature rich Debian based Linux Distribution that was created with various offensive and defensive hacking tools built in. While it excels in hacking related actives, it does lack by default some of the functionality that a person might want as their ‘daily driver’.
    - Source: https://www.kali.org/
- Ubuntu: Ubuntu is one to most beloved and used distributions of Linux. Not only does the ‘user based distribution’ very powerful and feature rich for the use a ‘daily driver’, but the ‘server based distribution’ is very feature rich and powerful as well. This is @c3rkah’s personal choice ‘daily driver’ because of the vast number of drivers that are built in and work right out of the box.
    - Source: https://ubuntu.com/
- Pop!_OS: Pop!_OS was built for STEM and creative professionals. They have built in tools for the topics of ‘Deep Learning’, ‘Engineering’, ‘Media Production’, and ‘Bioinformatics’. When @MicahKBrown does use Linux as his ‘daily driver’ this is his choice as the default skin / user interface is heavily inspired by MacOS.
    - Source: https://pop.system76.com/

‘Hey team, @c3rkah here NEVER, AND I MEAN EEEVVVVVVVVVVVVER talk to @MicahKBrown about Arch Linux!’

Take a moment to review these three options and select which version you want to use for the rest of this lab! Now we need to think about how are we going to run our Linux.

- Bear Metal: One of the cooler things about Linux is that most distributions are designed to run very well on older / legacy hardware. We have both taken legacy hardware that was struggling to run the bloats of other Oses and transformed them into perfectly suitable machines. If you do want a bear metal machine you should have no problem buying resold systems that have been phased out of use at other places as your Linux playground.
- Virtual Machine: A quicker way is to leverage a VM Platform. @MicahKBrown suggests Virtual Box as it is also Open Source and free to use for you.
- Cloud: You could choose to spin up a VM in the cloud if you want but how you do that different cloud platform to cloud platform.

Each of the listed Linux distributions has very good documentation on walking you through the install process that QCC will defer to as that documentation will change of time.

Instructions to install Ubuntu as a VM and load Kali Linux can be found at the end of this guide.

General Linux File:

It is important to remember that Linux is Open Source. That means the creator of the Distribution can change / modify all aspects of that distribution. Thus, things on your end might change.

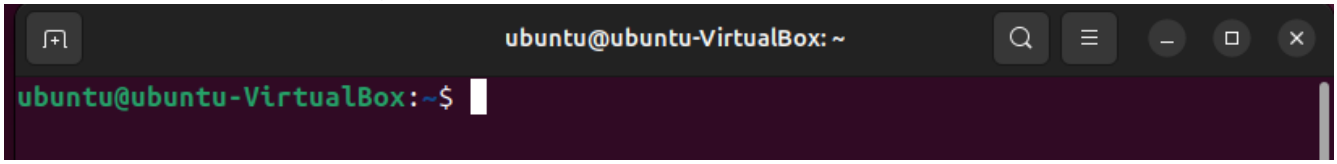
Many linux platforms implements the idea of separation of duties by initially only giving your actively logged on account general user permissions and requiring you to pre-pend the 'sudo' command before a command if you need 'admin' permissions.

It is also very important to remember that Linux is case sensitive. Thus the command 'ls' (or any other will not run if not typed exactly. This is one of the things that annoy those that are new to Linux.

In open transparency I am using Ubuntu Linux to get this labs screenshots.

In general the root or top most level of a Linux File system is called '/'. Under this object / container, all other files and folders are stored. Log into you Linux and open up the terminal.

1. From the command line will often give you contextual clues.



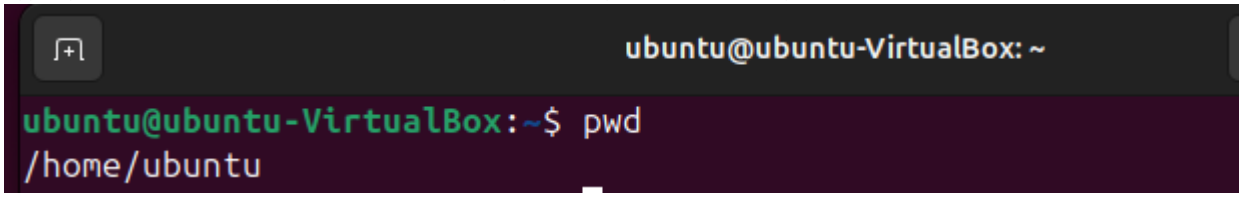
   a. Here you can see many things from left to right:

      i. My username is 'ubuntu'

      ii. My virtual machine name is ubuntu-VirtualBox

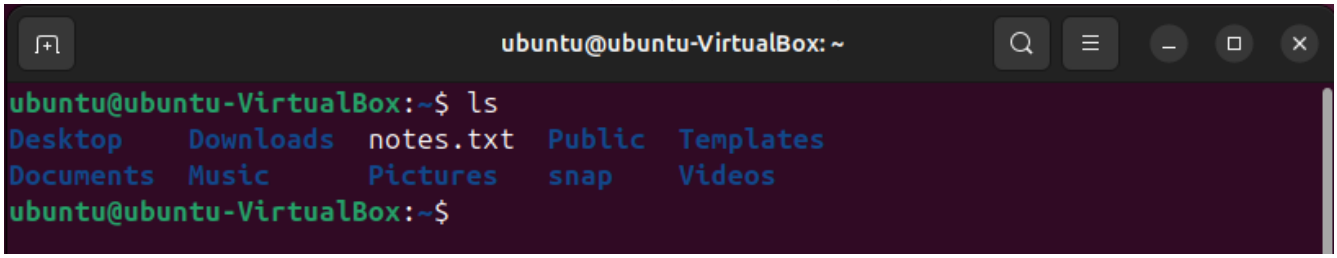
      iii. My 'present working directory' is set to '~'

2. Type 'pwd' to see where your 'present working directory' is located and press enter



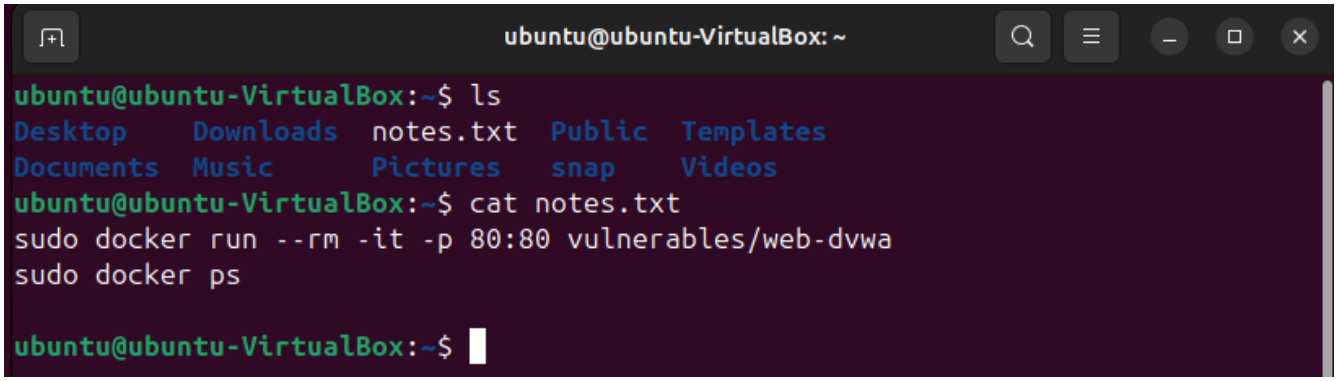
   a. Here you can see that under the '/' container we have a folder named 'home' that has a sub-folder for the 'ubuntu' account!

3. Type in the command ‘ls’ and hit enter to see what is in your ‘present working directory’



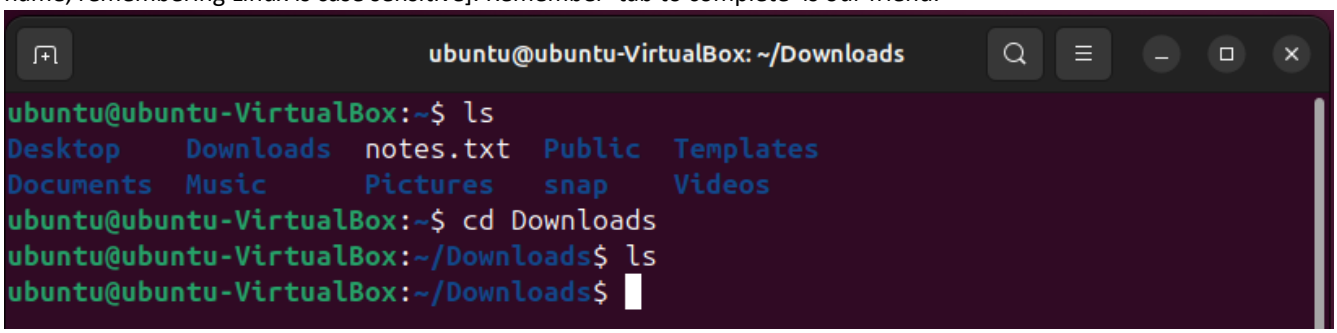
   a. Here we can see a lot of information! We can see a bunch of blue words, these are folders and we can see a white object ‘notes.txt’ which you guessed is a text file. The coloring might change version to version so play around or consult the friendly documentation.

4. Let’s quicly see what is in our text file by typing ‘cat notes.txt’ and pressing enter to list to contents of the referenced text file. It is important to remember that Linux is has a smart ‘auto complete’ feature so we could also have just typed ‘cat no’ and pressed tab or even ‘cat n’ and tab since no other objects start with ‘n’.



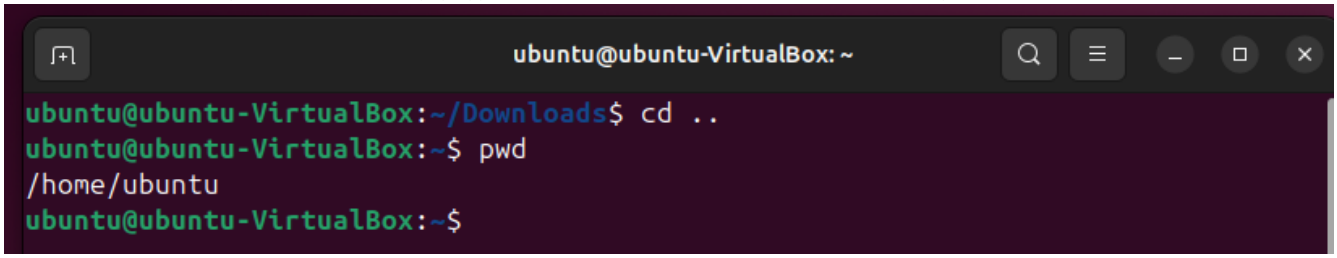
*You can ignore this for now if you do not have any files but trust me it works!

5. So now let’s explore moving around folders in the command line. To move into a folder simply type ‘cd [folder name, remembering Linux is case sensitive]. Remember ‘tab to complete’ is our friend.



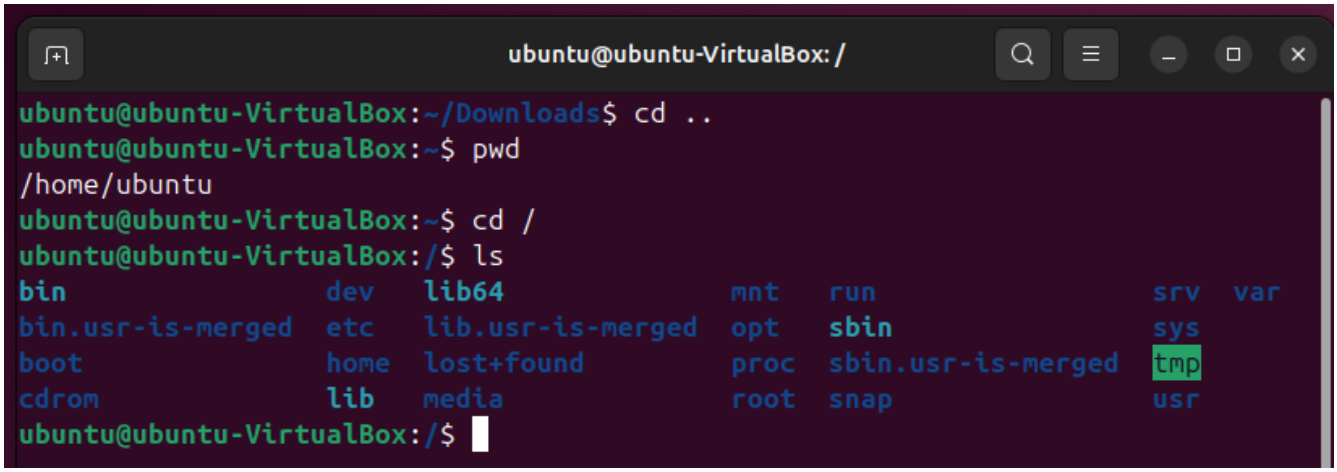
You now see my prompt has ben updated to show I am in the ‘~/Downloads’ folder.

6. To move back to the parent folder, simply type ‘cd ..’ and hit enter to move back to the parent folder.



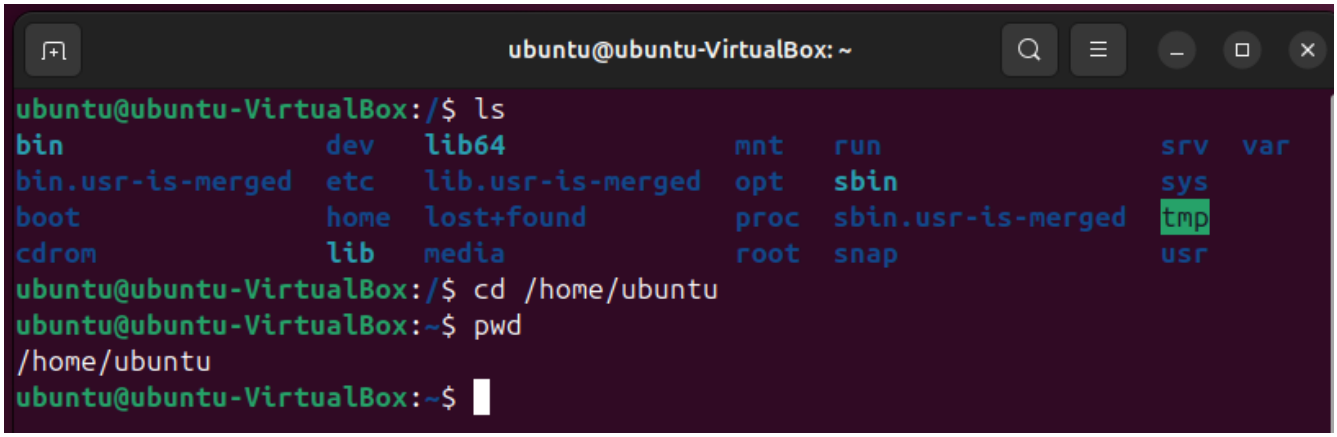
You can see that I am now back to my ‘/home/ubuntu’ folder!

7. You can also use the ‘cd’ command to move directly to any folder you just have to spell out the full path. So let’s go to the top of our file structure with ‘cd /’ and hit enter.

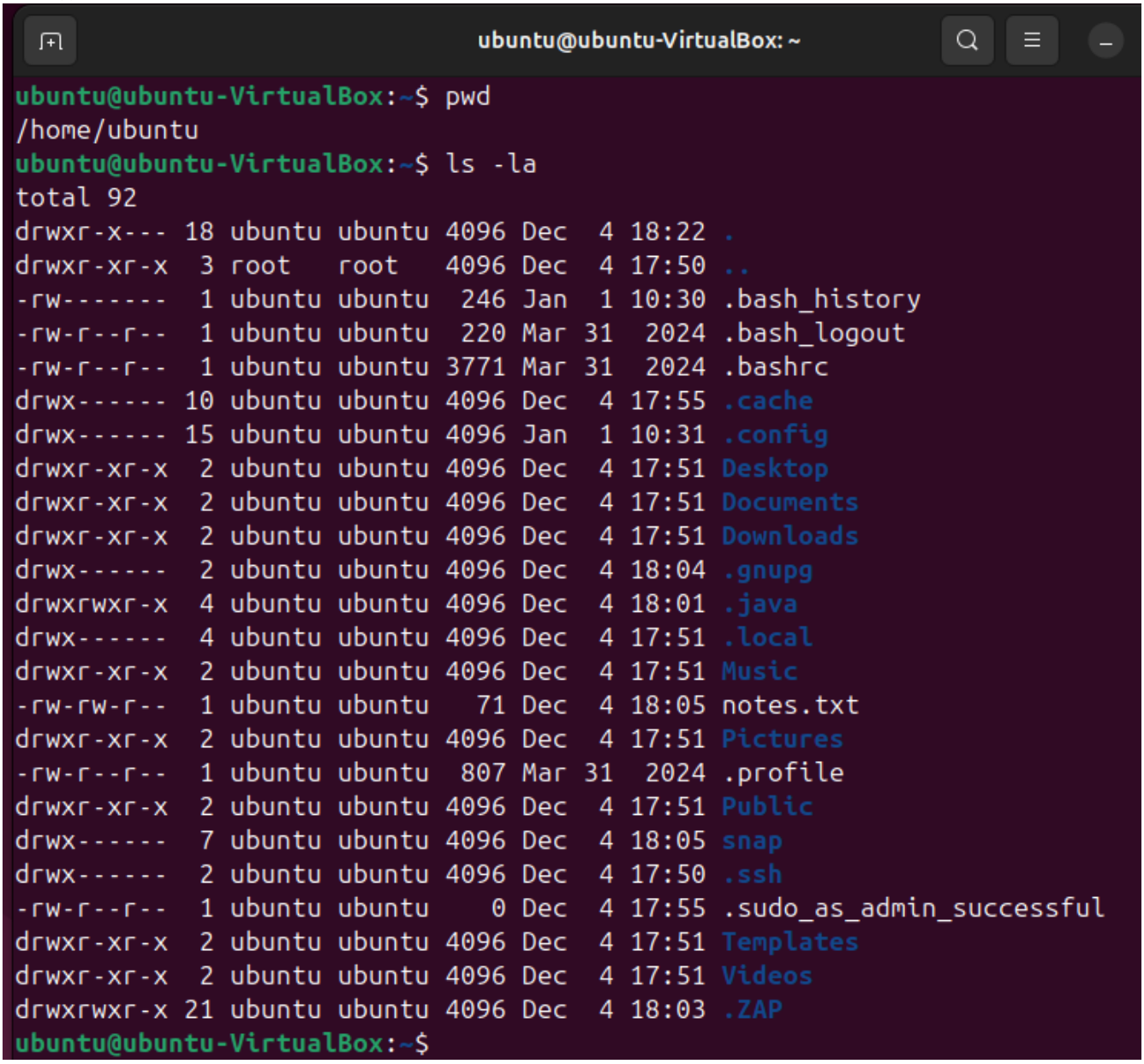


Thunder and lighting, very very frighting to me, to me!

8. Lets get back to our home on this system with ‘cd /home/ubuntu’



9. Now type ‘ls -la’ and hit enter



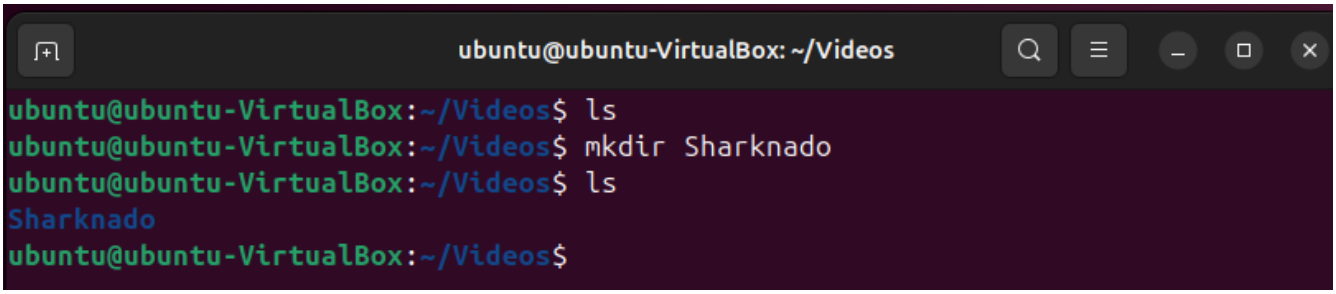
This gives us a new view of things where:

   a. The ‘-a’ variable shows all files (even those that start with a ‘.’ or are hidden.

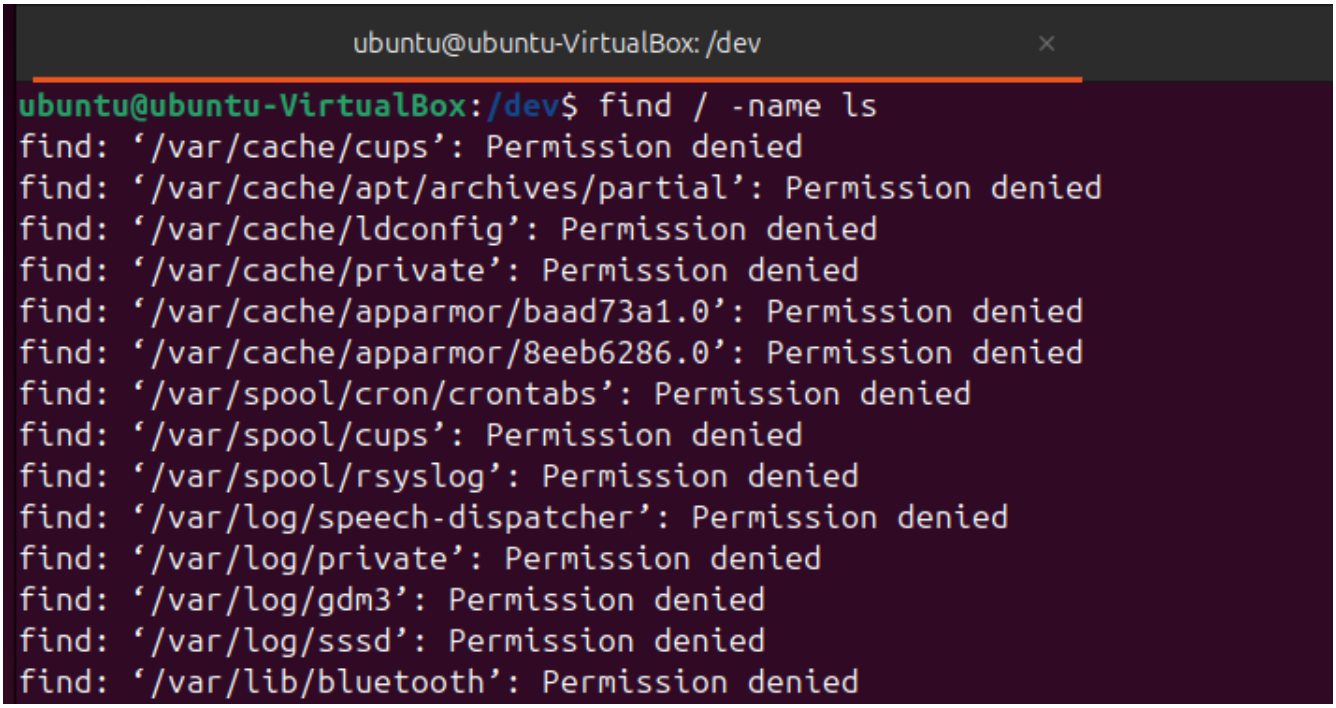
   b. The ‘-l’ variable shows additional detais such as permissions, owner, last updated.

10. Good news team!!! I have just purchase Sharknado on [Video Service Redacted] but need a place to save it on my Linux Laptopfor an upcoming flight. What are we do? Lets go into our Downloads folder with ‘cd Videos’ and enter. As you can see with a simple ‘ls’ and enter the folder is empty. Now I just need to make a folder for Sharknado. Type ‘mkdire Sharknado’ and hit enter. You can confirm with a simple ‘ls’ and enter that we now have

a folder to download our movie to!



11. At times we may need to search the filesystem to find where a file or application is. We can use the find command, type 'find / -name ls' to view where ls lives.



If you system gives you an error that a local database does not exist, you can simply run the 'updatedb' command (please note you might need to use the sudo command at the start depending on your Linux distribution.

12. Now lets explore the /etc/passwd file with ‘cat /etc/passwd’ and pressing enter

```
ubuntu@ubuntu-VirtualBox:~$ ls
Desktop    Downloads  notes.txt  Public  Templates
Documents  Music      Pictures   snap    Videos
ubuntu@ubuntu-VirtualBox:~$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
```

Note that you can specify files in other locations simply by providing the full path.

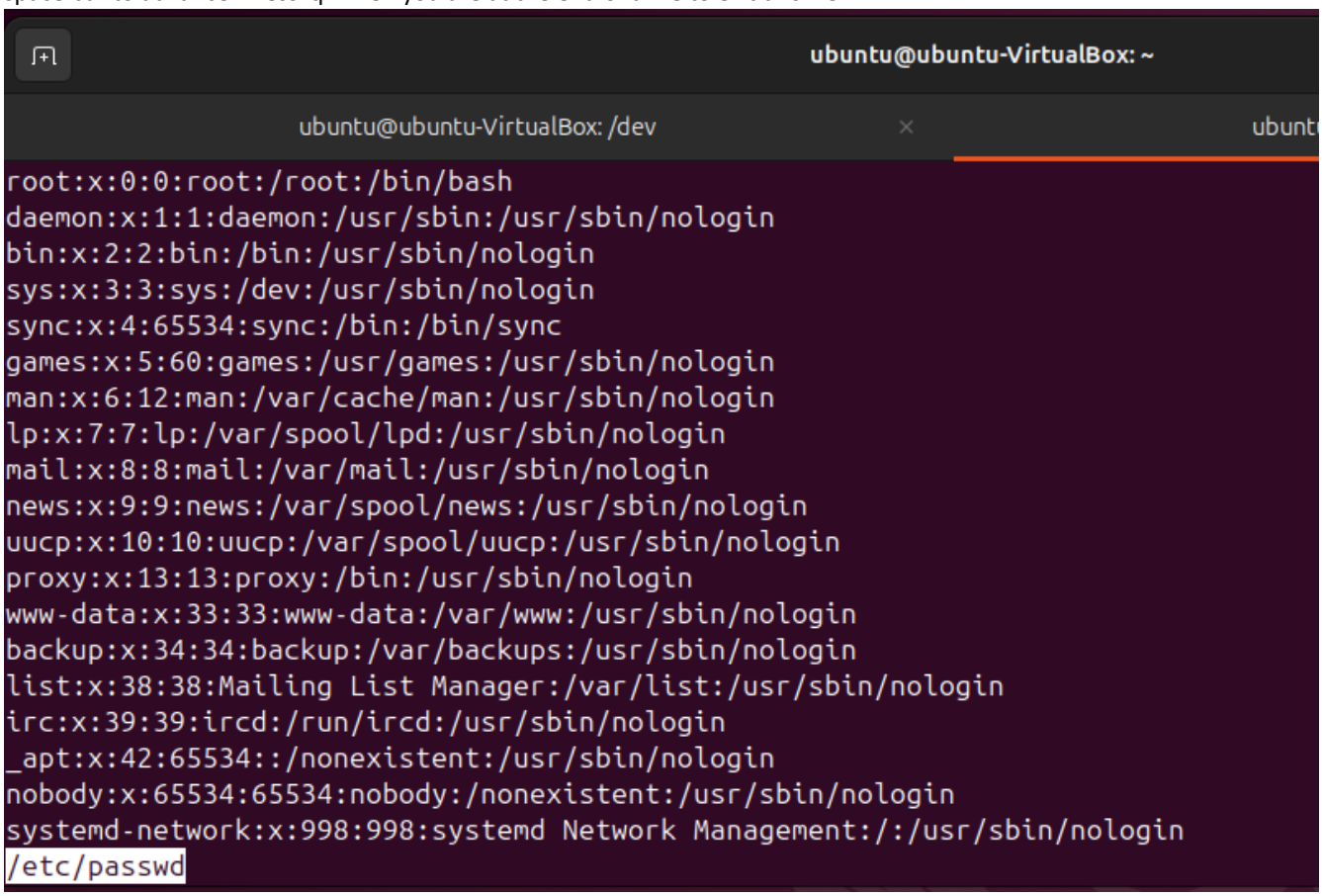
13. Now try ‘head /etc/passwd’ and enter which will only show the top of the file.

```
ubuntu@ubuntu-VirtualBox:~$ head /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
ubuntu@ubuntu-VirtualBox:~$
```

14. Now try ‘tail /etc/passwd’ and enter which will only show the end of the file.

```
ubuntu@ubuntu-VirtualBox:~$ tail /etc/passwd
cups-browsed:x:115:114::/nonexistent:/usr/sbin/nologin
hplip:x:116:7:HPLIP system user,,,:/run/hplip:/bin/false
gnome-remote-desktop:x:988:988:GNOME Remote Desktop:/var/lib/gnome-remote-desktop:/usr/sbin/nologin
polkitd:x:987:987:User for polkitd:/:/usr/sbin/nologin
rtkit:x:117:119:RealtimeKit,,,:/proc:/usr/sbin/nologin
colord:x:118:120:colord colour management daemon,,,:/var/lib/colord:/usr/sbin/nologin
gnome-initial-setup:x:119:65534::/run/gnome-initial-setup/:/bin/false
gdm:x:120:121:Gnome Display Manager:/var/lib/gdm3:/bin/false
nm-openvpn:x:121:122:NetworkManager OpenVPN,,,:/var/lib/openvpn/chroot:/usr/sbin/nologin
ubuntu:x:1000:1000:ubuntu:/home/ubuntu:/bin/bash
ubuntu@ubuntu-VirtualBox:~$
```

15. Now try ‘less /etc/passwd’ and enter which will allow you to return only one screen of a file making you pres the space bar to advance. Press ‘q’ when you are at the end of a file to exit this view.



Ok, so we have played around in the command line a bit! I hope you had fun! But let’s talk about many of the common objects you will find in many common Linux distributions knowing that this will very from Linux distribution to Linux distribution.

- ‘/’ this is the top most container that stores all other objects (files and folders)
    - ‘/dev’ stores devices (drivers terminals, etc)
    - ‘/etc’ holds configuration items. (Very fun to play with password cracking)
    - ‘/home’ contains the user’s home directories

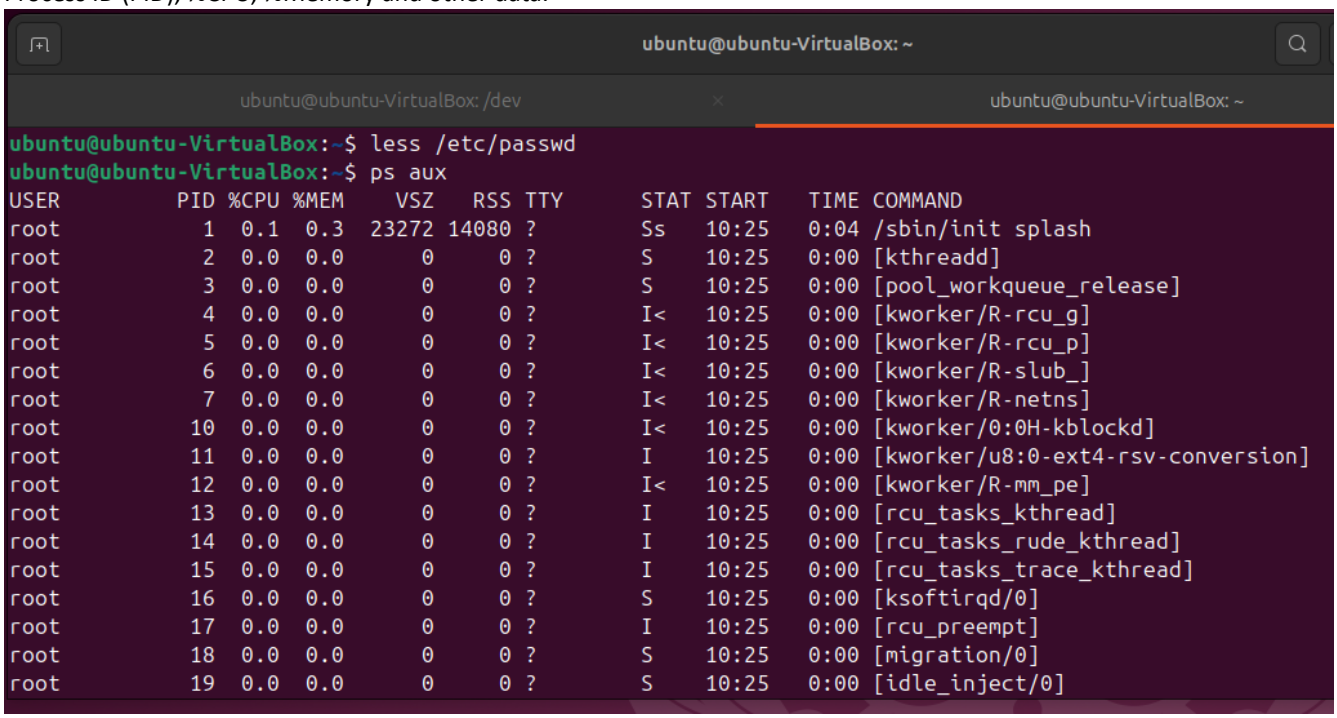
- ‘/lib’ contains common libraries
- ‘/mnt’ contains various remote and temporary file systems (DVD, thumb drives, shares, etc)
- ‘/opt’ store optional items (tools specific for the distrubution
- ‘/proc’ virtual file system that contains the Linux kernel
- ‘/root’ contains the rot (admin) account log on directory
- ‘/temp’ for temporary data and is cleared out at reboot
- ‘/usr’ contains user programs and other data’
- ‘/var’ holds many varios files including the log files

Take your time and explore your local set up to understand the file system better

Running Programs

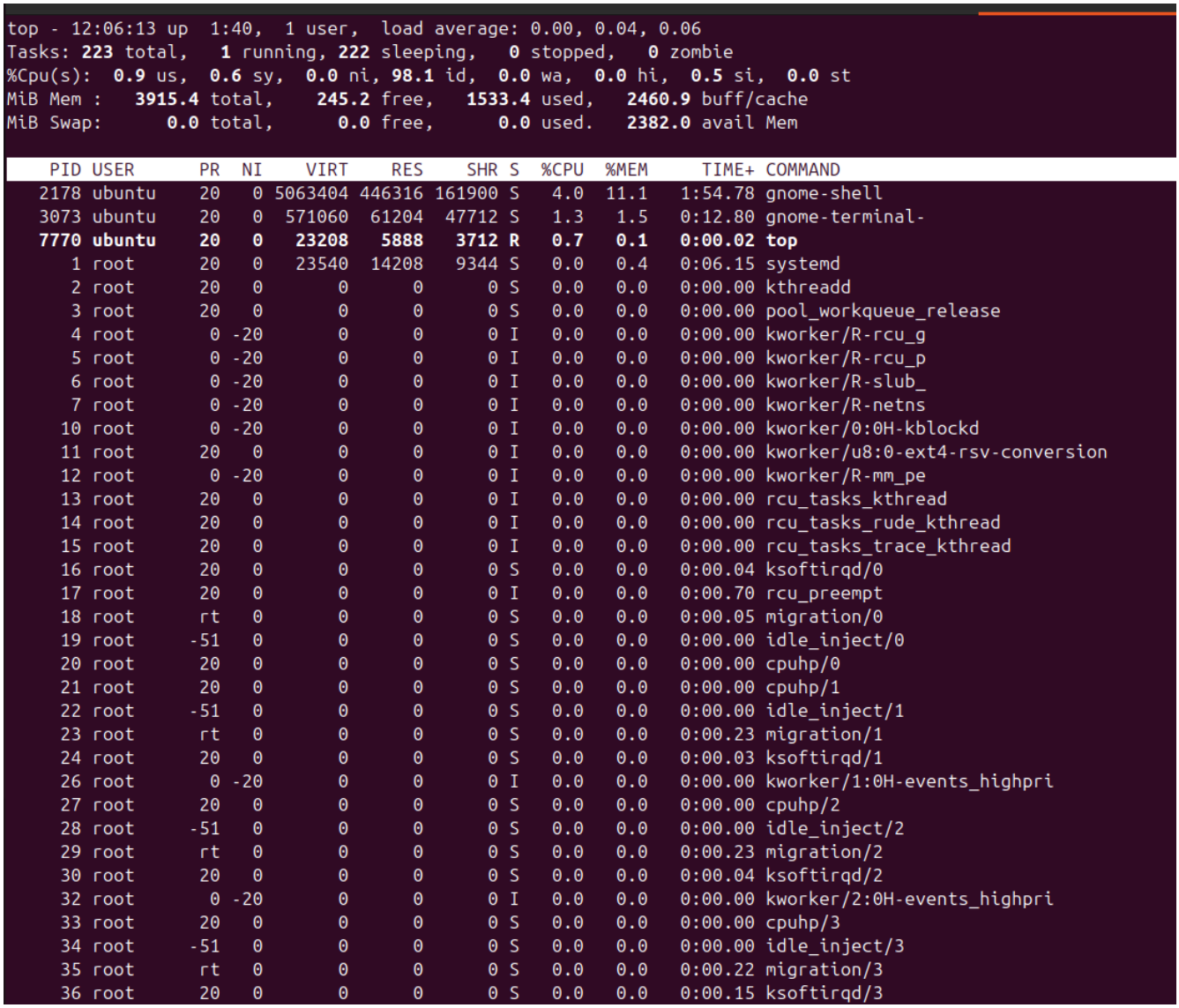
Now we are going to talk about running programs

1. In your command line type ‘ps aux’ to see what programs are running. This is nice as you can get inromation like Process ID (PID), %CPU, %Memory and other data.



You can use the ‘less’ trick to advance page by page with ‘ps aux | less’. Just remember to press ‘q’ at the end to escape!

2. ‘top’ is another very useful tool that will give you real time stats for applications and processes that are running. Just remember to press ‘q’ to exit.



```
top - 12:06:13 up  1:40,  1 user,  load average: 0.00, 0.04, 0.06
Tasks: 223 total,   1 running, 222 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.9 us,  0.6 sy,  0.0 ni, 98.1 id,  0.0 wa,  0.0 hi,  0.5 si,  0.0 st
MiB Mem :   3915.4 total,    245.2 free,   1533.4 used,   2460.9 buff/cache
MiB Swap:      0.0 total,      0.0 free,      0.0 used.   2382.0 avail Mem

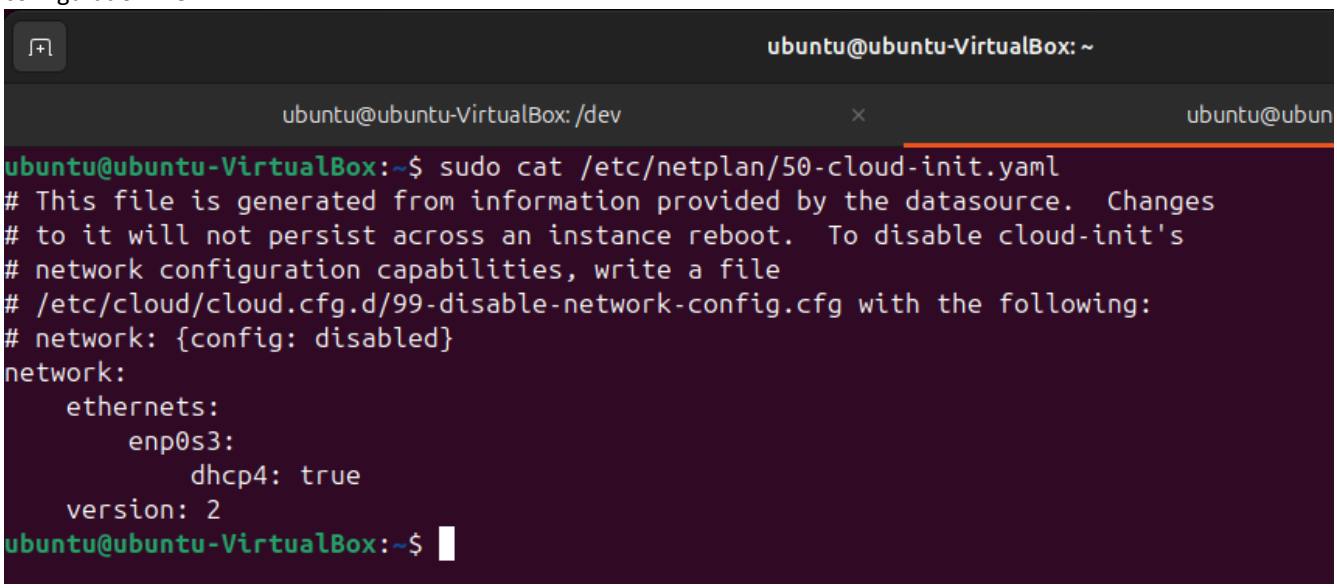
    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   2178 ubuntu    20   0 5063404 446316 161900 S   4.0  11.1   1:54.78 gnome-shell
   3073 ubuntu    20   0  571060  61204  47712 S   1.3   1.5   0:12.80 gnome-terminal-
   7770 ubuntu    20   0   23208   5888   3712 R   0.7   0.1   0:00.02 top
      1 root      20   0   23540  14208   9344 S   0.0   0.4   0:06.15 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kthreadd
      3 root      20   0       0      0      0 S   0.0   0.0   0:00.00 pool_workqueue_release
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-rcu_g
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-rcu_p
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-slub_
      7 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-netns
     10 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/0:0H-kblockd
     11 root      20   0       0      0      0 I   0.0   0.0   0:00.00 kworker/u8:0-ext4-rsv-conversion
     12 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-mm_pe
     13 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_kthread
     14 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_rude_kthread
     15 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_trace_kthread
     16 root      20   0       0      0      0 S   0.0   0.0   0:00.04 ksoftirqd/0
     17 root      20   0       0      0      0 I   0.0   0.0   0:00.70 rcu_preempt
     18 root      rt   0       0      0      0 S   0.0   0.0   0:00.05 migration/0
     19 root     -51   0       0      0      0 S   0.0   0.0   0:00.00 idle_inject/0
     20 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/0
     21 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/1
     22 root     -51   0       0      0      0 S   0.0   0.0   0:00.00 idle_inject/1
     23 root      rt   0       0      0      0 S   0.0   0.0   0:00.23 migration/1
     24 root      20   0       0      0      0 S   0.0   0.0   0:00.03 ksoftirqd/1
     26 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/1:0H-events_highpri
     27 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/2
     28 root     -51   0       0      0      0 S   0.0   0.0   0:00.00 idle_inject/2
     29 root      rt   0       0      0      0 S   0.0   0.0   0:00.23 migration/2
     30 root      20   0       0      0      0 S   0.0   0.0   0:00.04 ksoftirqd/2
     32 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/2:0H-events_highpri
     33 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/3
     34 root     -51   0       0      0      0 S   0.0   0.0   0:00.00 idle_inject/3
     35 root      rt   0       0      0      0 S   0.0   0.0   0:00.22 migration/3
     36 root      20   0       0      0      0 S   0.0   0.0   0:00.15 ksoftirqd/3
```

3. Sometimes you need to find where a program lives. To do this use the ‘find’ command. In your command line, type ‘find / -name ls’ and hit enter.

```
ubuntu@ubuntu-VirtualBox:~$ fine / -name ls
Command 'fine' not found, did you mean:
  command 'fiwe' from snap fiwe (latest)
  command 'ifne' from deb moreutils (0.67-1)
  command 'find' from deb findutils (4.9.0-5)
  command 'wine' from deb wine (8.0.1~repack-3ubuntu2)
  command 'file' from deb file (1:5.45-2)
  command 'xine' from deb xine-ui (0.99.14-1)
See 'snap info <snapname>' for additional versions.
ubuntu@ubuntu-VirtualBox:~$
```

Networking

1. From your command line type ‘sudo cat /etc/netplan/50-cloud0init.yaml’ and hit enter to see your networking configuration file.



Other versions of Linux traditionally have had this in /etc/network/interfaces/

2. This is great if we want to view it but if we want to change it we need an actual editor. I find nano to be the superior text editor. Type ‘sudo nano /etc/netplan/50-cloud0init.yaml’

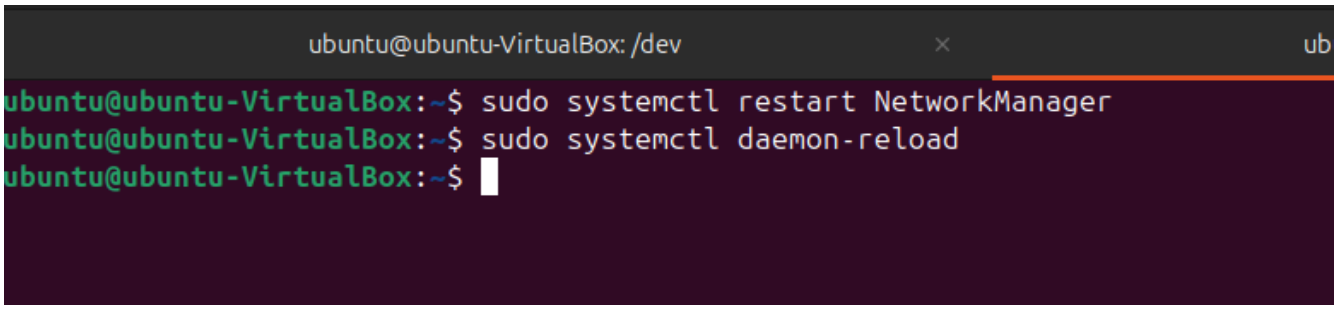


You can make any edits you want here. Lines that start with '#' are commented out. So just add your initials or the word of swear o your choice to the comment line of your choice and then press 'Control + X' to exit. As nano sees you made a change it will ask you do you want to save the change press Y and then hit enter as you want to make the change to that file. You will be now exited.

3. Type 'sudo cat /etc/netplan/50-cloud0init.yaml' to make sure your changes were added.

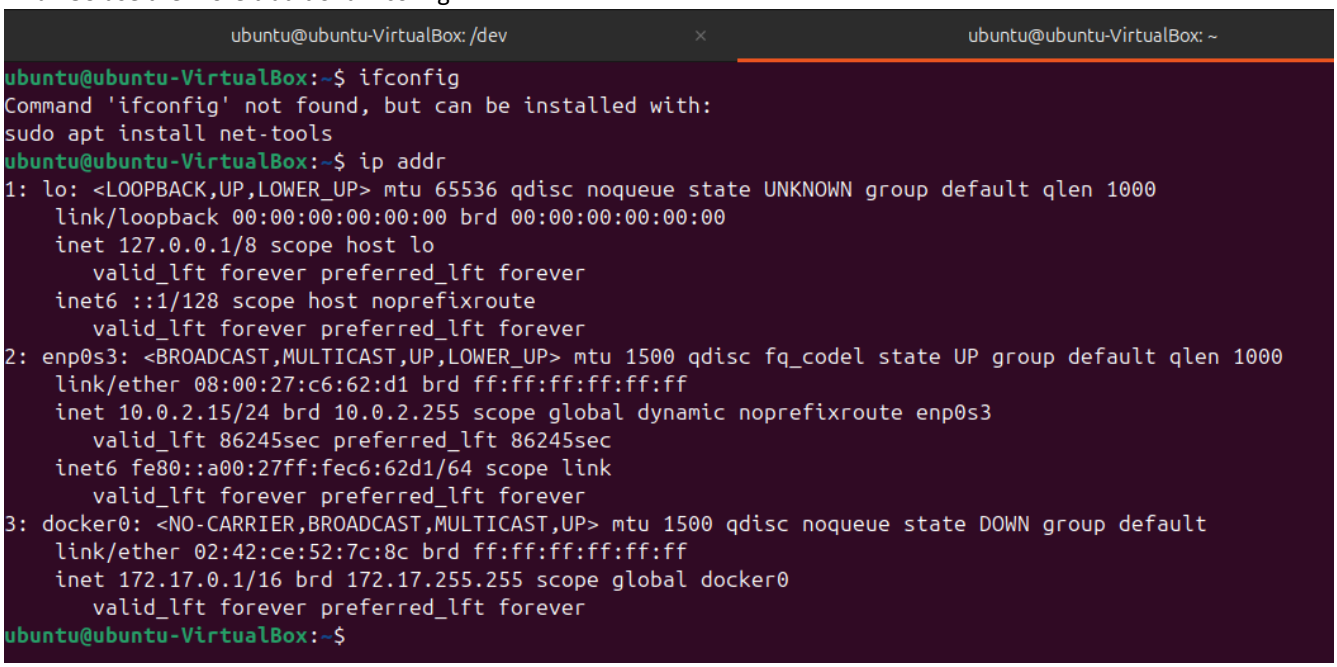


4. You can always restart your networking services with 'sudo systemctl restart NetworkManager' the additional command 'sudo systemctl daemon-reload' may be needed to restart networking services.

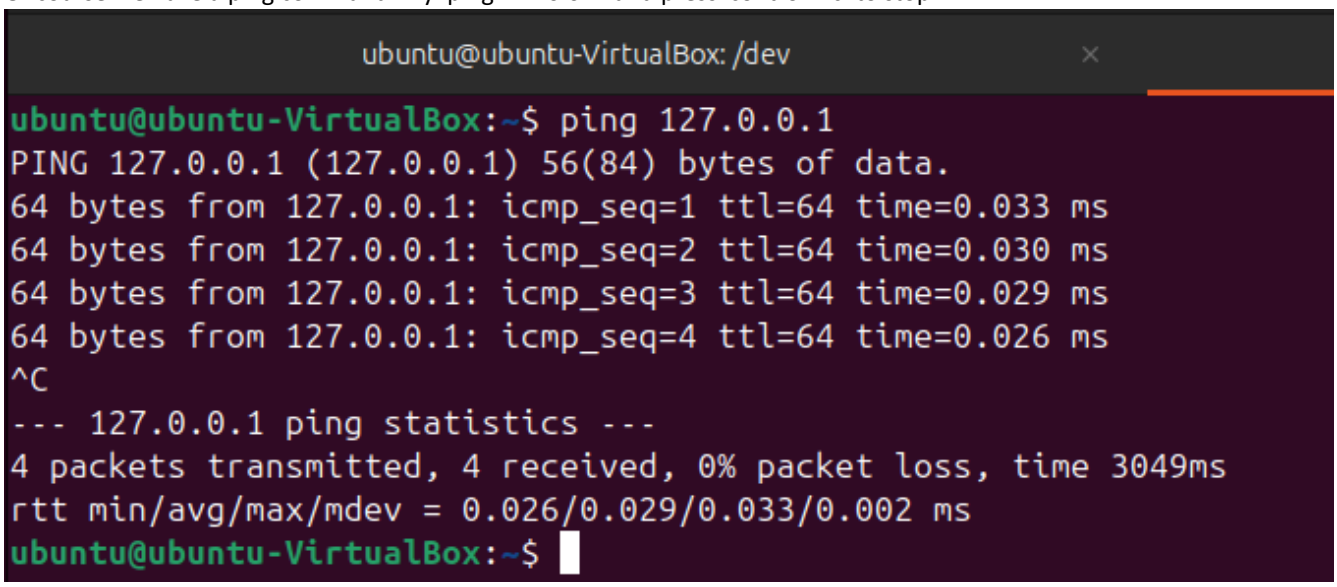


Hi friends, @MicahKBrown here, other more civilized Linux OS might simply need the 'service networking restart' command to do this. Until Next time!

5. To see the actual ip configuration on your system in Ubuntu use the 'ip addr' command in Ubuntu. More civilized Linux OS use the more traditional ifconfig.



6. Of course we have a ping command. Try 'ping 127.0.0.1' and press 'control + c' to stop.



7. To see netstat statistics we need to install it with ‘sudo apt install net-tools’ if it is not installed (like on Ubuntu)

```
ubuntu@ubuntu-VirtualBox:~$ netstat -nap
Command 'netstat' not found, but can be installed with:
sudo apt install net-tools
ubuntu@ubuntu-VirtualBox:~$ sudo apt install net-tools
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  net-tools
0 upgraded, 1 newly installed, 0 to remove and 87 not upgraded.
Need to get 204 kB of archives.
After this operation, 811 kB of additional disk space will be used.
Get:1 http://us.archive.ubuntu.com/ubuntu noble/main amd64 net-tools amd64 2.10-0.1ubuntu4 [204 kB]
Fetched 204 kB in 0s (522 kB/s)
Selecting previously unselected package net-tools.
(Reading database ... 149650 files and directories currently installed.)
Preparing to unpack .../net-tools_2.10-0.1ubuntu4_amd64.deb ...
Unpacking net-tools (2.10-0.1ubuntu4) ...
Setting up net-tools (2.10-0.1ubuntu4) ...
Processing triggers for man-db (2.12.0-4build2) ...
ubuntu@ubuntu-VirtualBox:~$
```

8. Use the ‘netstat -nap’ command to see your active network connections / ports.

```
ubuntu@ubuntu-VirtualBox:~$ netstat -nap
(Not all processes could be identified, non-owned process info
 will not be shown, you would have to be root to see it all.)
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 127.0.0.53:53           0.0.0.0:*               LISTEN      -
tcp        0      0 127.0.0.1:32997         0.0.0.0:*               LISTEN      -
tcp        0      0 127.0.0.1:631           0.0.0.0:*               LISTEN      -
tcp        0      0 127.0.0.54:53           0.0.0.0:*               LISTEN      -
tcp        0      0 10.0.2.15:59786         91.189.91.81:80         TIME_WAIT   -
tcp        0      0 10.0.2.15:47530         142.251.179.101:443     TIME_WAIT   -
tcp        0      0 10.0.2.15:59196         35.201.103.21:443       TIME_WAIT   -
tcp        0      0 10.0.2.15:41266         35.190.72.216:443       TIME_WAIT   -
tcp        0      0 10.0.2.15:53896         34.107.243.93:443       ESTABLISHED 6435/firefox
tcp        0      0 10.0.2.15:50930         34.160.90.233:443       TIME_WAIT   -
tcp        0      0 10.0.2.15:41260         35.190.72.216:443       TIME_WAIT   -
tcp        0      0 10.0.2.15:41396         34.117.121.53:443       TIME_WAIT   -
tcp        0      0 10.0.2.15:55360         35.244.181.201:443      TIME_WAIT   -
tcp        0      0 10.0.2.15:41260         34.149.100.209:443      TIME_WAIT   -
tcp        0      0 10.0.2.15:36356         172.64.41.4:443         ESTABLISHED 6435/firefox
tcp        0      0 10.0.2.15:38118         34.160.144.191:443      TIME_WAIT   -
tcp6       0      0 ::1:631                 :::*                    LISTEN      -
udp        0      0 0.0.0.0:37399           0.0.0.0:*                           -
udp        0      0 0.0.0.0:5353            0.0.0.0:*                           -
udp        0      0 127.0.0.54:53           0.0.0.0:*                           -
udp        0      0 127.0.0.53:53           0.0.0.0:*                           -
udp        0      0 10.0.2.15:68            10.0.2.2:67             ESTABLISHED -
udp6       0      0 :::53827                :::*                                -
udp6       0      0 :::5353                 :::*                                -
Active UNIX domain sockets (servers and established)
Proto RefCnt Flags       Type       State         I-Node   PID/Program name     Path
unix  3      [ ]         STREAM     CONNECTED     37360    6435/firefox
unix  3      [ ]         STREAM     CONNECTED     15267    -                    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     15159    -                    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     14728    -                    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     8911     -
unix  3      [ ]         STREAM     CONNECTED     16415    2136/gnome-session-
unix  3      [ ]         STREAM     CONNECTED     10041    -                    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     18469    2192/dbus-daemon     /run/user/1000/at-spi/bus
unix  3      [ ]         STREAM     CONNECTED     15137    2297/gsd-datetime
unix  3      [ ]         STREAM     CONNECTED     6914     -                    /run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     15018    -                    /run/systemd/journal/stdout
```

9. This is a lot to take in so let's use our 'less' trick and type in 'netstat -nap | less'

```
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 127.0.0.53:53           0.0.0.0:*               LISTEN      -
tcp        0      0 127.0.0.1:32997         0.0.0.0:*               LISTEN      -
tcp        0      0 127.0.0.1:631           0.0.0.0:*               LISTEN      -
tcp        0      0 127.0.0.54:53           0.0.0.0:*               LISTEN      -
tcp        0      0 10.0.2.15:53896         34.107.243.93:443       ESTABLISHED 6435/firefox
tcp        0      0 10.0.2.15:43870         142.251.166.113:443     ESTABLISHED 6435/firefox
tcp        0      0 10.0.2.15:46442         172.253.63.147:443      ESTABLISHED 6435/firefox
tcp        0      0 10.0.2.15:51522         35.244.181.201:443      ESTABLISHED 6435/firefox
tcp        0      0 10.0.2.15:58760         34.160.144.191:443      ESTABLISHED 6435/firefox
tcp        0      0 10.0.2.15:36356         172.64.41.4:443         ESTABLISHED 6435/firefox
tcp6       0      0 ::1:631                 :::*                    LISTEN      -
udp        0      0 0.0.0.0:37399           0.0.0.0:*                           -
udp        0      0 0.0.0.0:5353            0.0.0.0:*                           -
udp        0      0 127.0.0.54:53           0.0.0.0:*                           -
udp        0      0 127.0.0.53:53           0.0.0.0:*                           -
udp        0      0 10.0.2.15:68            10.0.2.2:67             ESTABLISHED -
udp6       0      0 :::53827                :::*                                -
udp6       0      0 :::5353                 :::*                                -
Active UNIX domain sockets (servers and established)
Proto RefCnt Flags       Type       State         I-Node   PID/Program name     Path
unix  3      [ ]         STREAM     CONNECTED     37360    6435/firefox
unix  3      [ ]         STREAM     CONNECTED     15267    -                    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     15159    -                    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     14728    -                    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     8911     -
unix  3      [ ]         STREAM     CONNECTED     16415    2136/gnome-session-
unix  3      [ ]         STREAM     CONNECTED     10041    -                    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     18469    2192/dbus-daemon     /run/user/1000/at-spi/bus
unix  3      [ ]         STREAM     CONNECTED     15137    2297/gsd-datetime
unix  3      [ ]         STREAM     CONNECTED     6914     -                    /run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     15018    -                    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     37376    6435/firefox
unix  3      [ ]         STREAM     CONNECTED     37377    7172/firefox
unix  3      [ ]         STREAM     CONNECTED     15275    2290/ibus-daemon
unix  3      [ ]         STREAM     CONNECTED     17495    2304/gsd-housekeepi
unix  3      [ ]         STREAM     CONNECTED     6962     -                    /run/dbus/system_bus_socket
unix  3      [ ]         DGRAM      CONNECTED     5319     -
unix  3      [ ]         STREAM     CONNECTED     18582    -                    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     18465    1954/dbus-daemon     /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     16590    2329/evolution-alar
```

In an active investigation it might be helpful to look for LISTENING and ESTABLISHED connections.

10. You can also use the greep to return only lines that match the following term for example lets try it out with LISTENING and ESTABLISHED as our search terms.

```
ubuntu@ubuntu-VirtualBox:~$ netstat -nap | grep ESTABLISHED
(Not all processes could be identified, non-owned process info
 will not be shown, you would have to be root to see it all.)
tcp        0      0 10.0.2.15:53896         34.107.243.93:443       ESTABLISHED 6435/firefox
tcp        0      0 10.0.2.15:38676         34.120.208.123:443      ESTABLISHED 6435/firefox
tcp        0      0 10.0.2.15:36356         172.64.41.4:443         ESTABLISHED 6435/firefox
udp        0      0 10.0.2.15:68            10.0.2.2:67             ESTABLISHED -
ubuntu@ubuntu-VirtualBox:~$ netstat -nap | grep LISTEN
(Not all processes could be identified, non-owned process info
 will not be shown, you would have to be root to see it all.)
tcp        0      0 127.0.0.53:53           0.0.0.0:*               LISTEN      -
tcp        0      0 127.0.0.1:32997         0.0.0.0:*               LISTEN      -
tcp        0      0 127.0.0.1:631           0.0.0.0:*               LISTEN      -
tcp        0      0 127.0.0.54:53           0.0.0.0:*               LISTEN      -
tcp6       0      0 ::1:631                 :::*                    LISTEN      -
unix  2      [ ACC ]     STREAM     LISTENING     401      -                    /run/systemd/io.systemd.sysext
unix  2      [ ACC ]     STREAM     LISTENING     13076    -                    /var/run/docker/libnetwork/06c8521a42f6.sock
unix  2      [ ACC ]     STREAM     LISTENING     11077    -                    /run/user/1000/systemd/private
unix  2      [ ACC ]     STREAM     LISTENING     11084    1954/dbus-daemon     /run/user/1000/bus
unix  2      [ ACC ]     STREAM     LISTENING     11086    -                    /run/user/1000/gnupg/S.dirmngr
unix  2      [ ACC ]     STREAM     LISTENING     11088    2097/gcr-ssh-agent   /run/user/1000/gcr/ssh
unix  2      [ ACC ]     STREAM     LISTENING     11090    1944/gnome-keyring-  /run/user/1000/keyring/control
unix  2      [ ACC ]     STREAM     LISTENING     11092    -                    /run/user/1000/gnupg/S.gpg-agent.browser
unix  2      [ ACC ]     STREAM     LISTENING     11094    -                    /run/user/1000/gnupg/S.gpg-agent.extra
unix  2      [ ACC ]     STREAM     LISTENING     11096    -                    /run/user/1000/gnupg/S.gpg-agent
unix  2      [ ACC ]     STREAM     LISTENING     11098    -                    /run/user/1000/gnupg/S.keyboxd
unix  2      [ ACC ]     STREAM     LISTENING     11100    1943/pipewire-pulse  /run/user/1000/pulse/native
unix  2      [ ACC ]     STREAM     LISTENING     11102    1932/pipewire        /run/user/1000/pipewire-0
unix  2      [ ACC ]     STREAM     LISTENING     11104    1932/pipewire        /run/user/1000/pipewire-0-manager
unix  2      [ ACC ]     STREAM     LISTENING     11106    -                    /run/user/1000/pk-debconf-socket
unix  2      [ ACC ]     STREAM     LISTENING     11108    -                    /run/user/1000/snapd-session-agent.socket
unix  2      [ ACC ]     STREAM     LISTENING     11110    -                    /run/user/1000/speech-dispatcher/speechd.sock
unix  2      [ ACC ]     STREAM     LISTENING     11118    -                    /run/user/1000/gnupg/S.gpg-agent.ssh
unix  2      [ ACC ]     STREAM     LISTENING     14136    1944/gnome-keyring-  /run/user/1000/keyring/pkcs11
unix  2      [ ACC ]     STREAM     LISTENING     14961    1944/gnome-keyring-  /run/user/1000/keyring/ssh
unix  2      [ ACC ]     STREAM     LISTENING     15850    2192/dbus-daemon     /run/user/1000/at-spi/bus
unix  2      [ ACC ]     STREAM     LISTENING     16512    2178/gnome-shell     /run/user/1000/wayland-0
unix  2      [ ACC ]     STREAM     LISTENING     15282    2290/ibus-daemon     /home/ubuntu/.cache/ibus/dbus-jrKcomY0
```

Please keep in mind grep works with many other Linux commands such as cat and ps. VERY useful

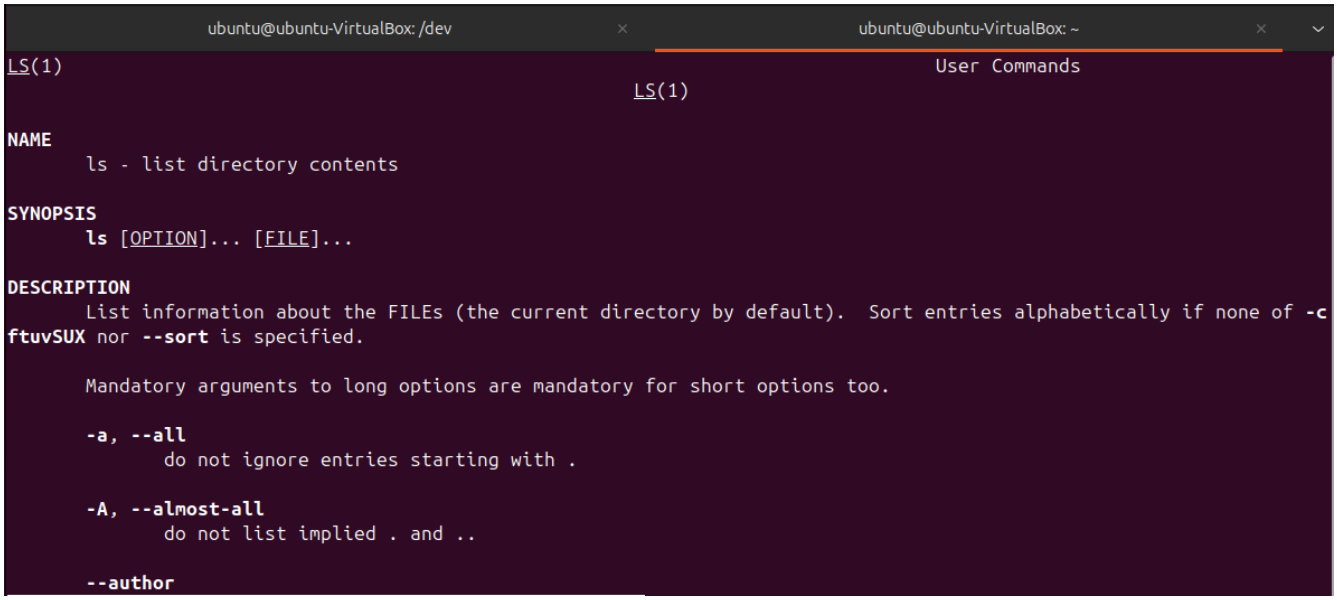
11. Likewise you can also use 'lsof -I | less' to further reduce information overflow

```
ubuntu@ubuntu-VirtualBox: /dev                                  ubuntu@ubuntu-VirtualBox

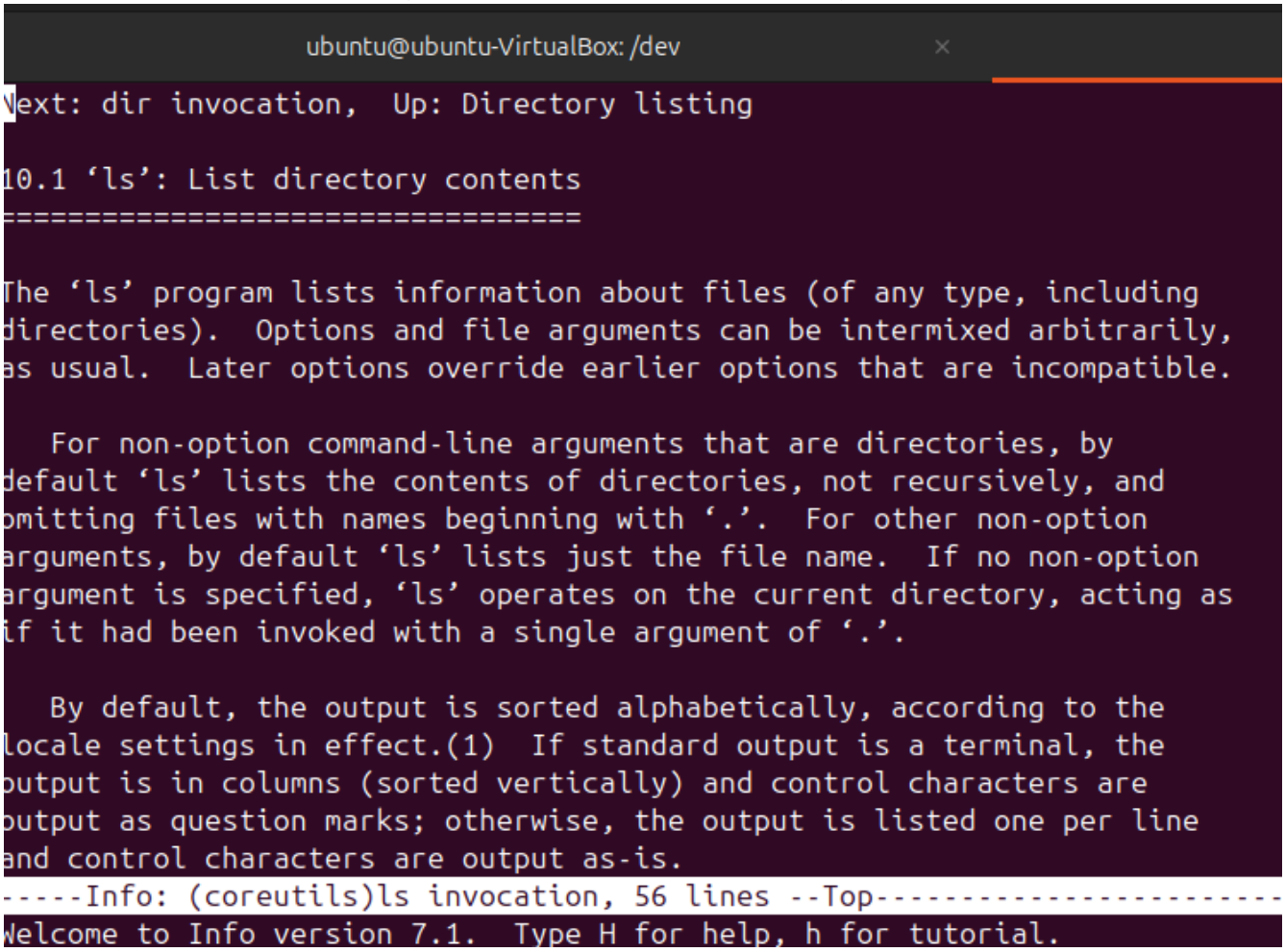
COMMAND  PID   USER   FD   TYPE DEVICE SIZE/OFF NODE NAME
firefox 6435 ubuntu   66u  IPv4  38662      0t0  TCP ubuntu-VirtualBox:51522->201.181.244.35.bc.googleusercontent.com:https (ESTABLISHED)
firefox 6435 ubuntu   98u  IPv4  38663      0t0  TCP ubuntu-VirtualBox:58760->191.144.160.34.bc.googleusercontent.com:https (ESTABLISHED)
firefox 6435 ubuntu  125u  IPv4  35189      0t0  TCP ubuntu-VirtualBox:36356->172.64.41.4:https (ESTABLISHED)
firefox 6435 ubuntu  151u  IPv4  36511      0t0  TCP ubuntu-VirtualBox:53896->93.243.107.34.bc.googleusercontent.com:https (ESTABLISHED)
(END)
```

Other useful tips and tricks

1. You can use the man command to read the manual for any executable that has a manual file defined. For example type ‘man ls’ to bring up the manual. Press ‘q’ to exit.



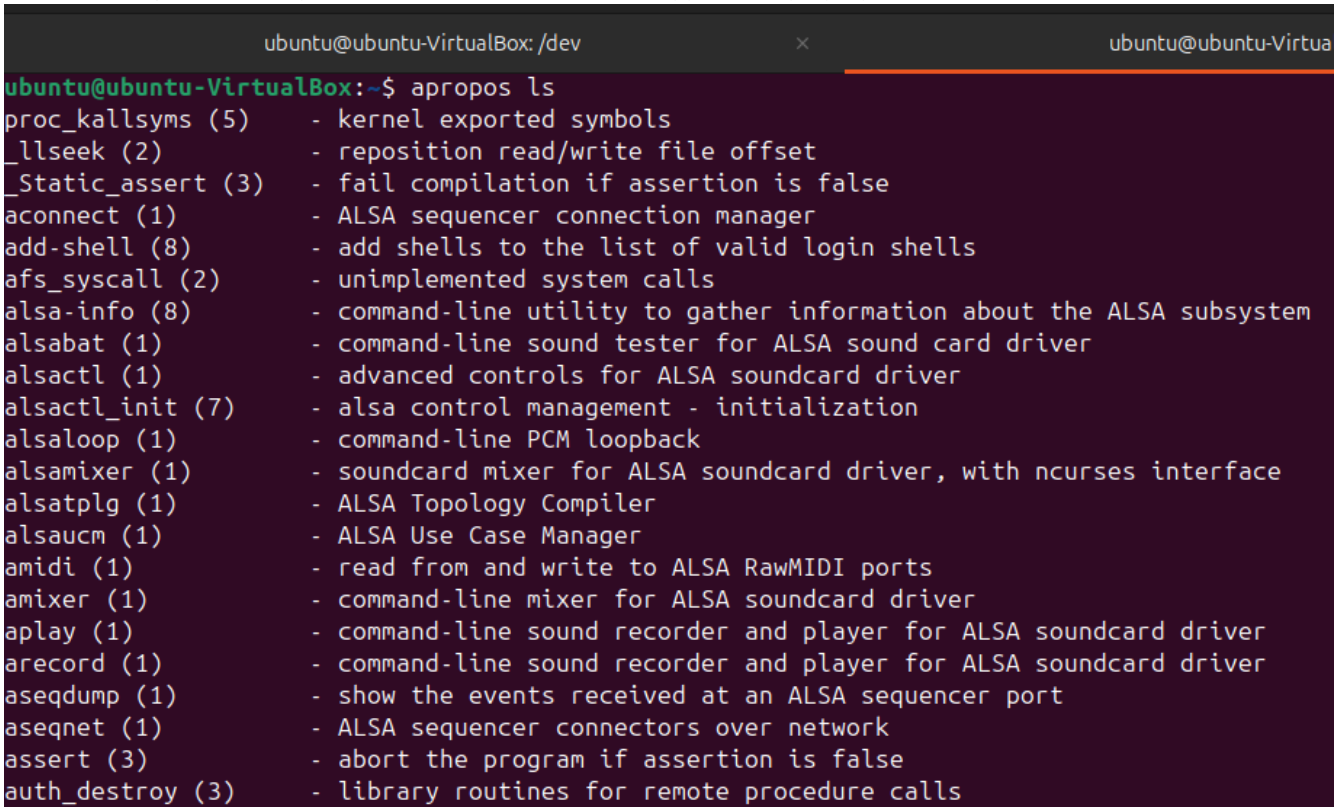
2. You can also use the info command to get more details on an executable. Let’s try ‘info ls’



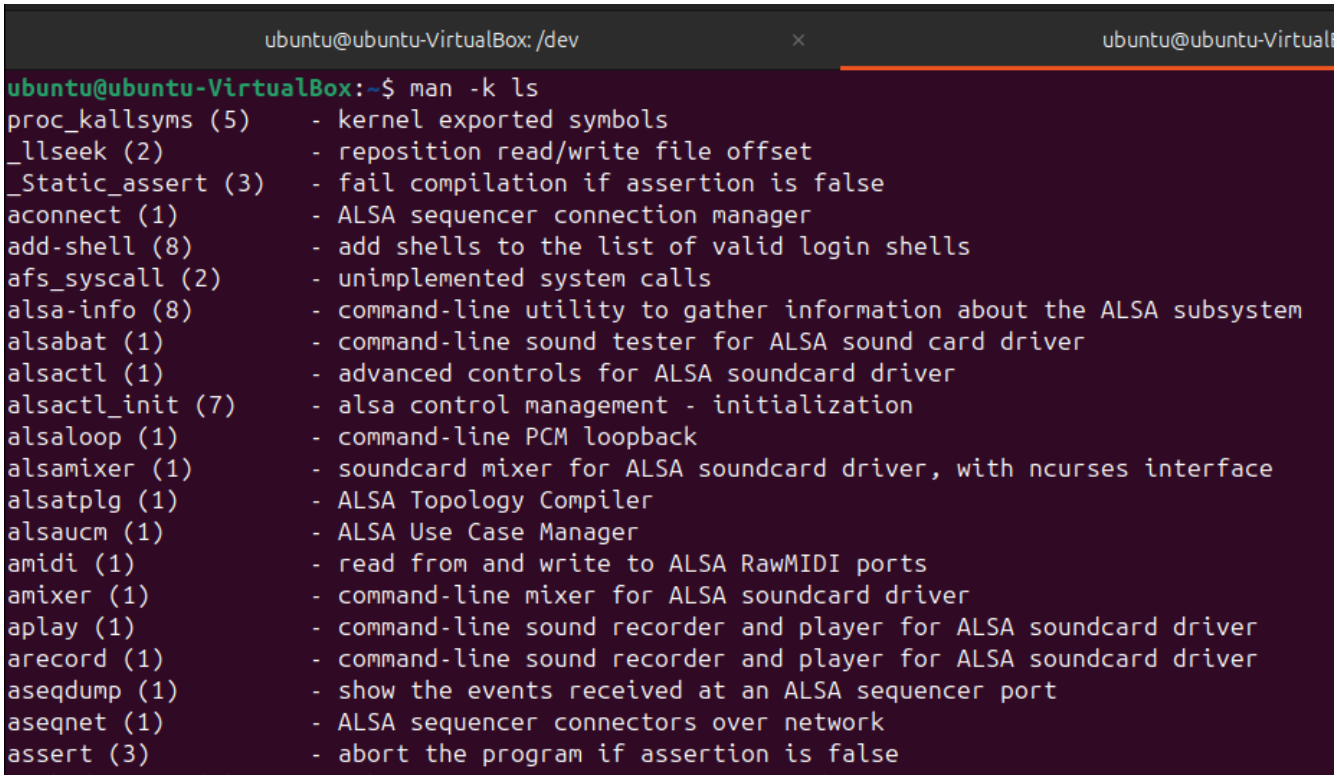
3. You can get summary information on an application with the whatis command. Let's try 'whatis ls'



4. You can even get a list of topics and commsnas related to an app. Let's try 'apropos ls'



5. You can even use the ‘man -k’ command to search defined manuals for the search term. Let’s try ‘man -k ls’



6. From the command line you can install applications by using ‘sudo apt-get install [application name]’
    a. I highly recommend video-downloader. It will download audio and video from many popular sites such as youtube and rumble! Great to prep for long trips. To install it type ‘sudo apt-get install video-downloader’
7. To shutdown a system type ‘sudo shutdown -h now’.
8. To reboot a system ‘sudo shutdown -r now’ or ‘reboot’
9. FINALLY! To upgrade out Linux you need two commands. Open up the terminal and type ‘sudo apt update’ and provide your password. This will update the repository list that your system is configured to use. Then enter the command ‘sudo apt upgrade’. Press ‘Y’ when prompted to install the latest version of software.

Friends remember Linux can be equally powerful as it is frustrating. Our recommendation is to try a few VMs to find your preferred distribution before you commit to loading that version on a physical machine.

Until Next Time!!!

Adenium:

Installing Kali Prebuilt VM

Kali Linux with VirtualBox as the Hypervisor

Download VirtualBox:

1. Go to: https://www.virtualbox.org/ and click downloads
2. Select your distribution to start your download
3. So much simpler than VMWare

Install Virtual Box:

1. Go to download location and double click. Accept UAC pop up.
2. Please note that exact install options might change as app is updated. This guide was created 8/24/2024.
3. On the first generic install screen click next.
4. The next screen allows you to customize your Virtual Box install. I kept with the default and clicked next.
5. Click Yes to the network interruption message.
6. Click yes if you get any missing dependencies.
7. At the next customization screen, I kept the default.
8. Click Install on the 'Ready to Install Screen'
9. Click Finish

Make sure that you are connected to the internet.

Now to this site and to download the ISO file for Ubuntu in the location of your choice:
https://ubuntu.com/download/desktop

Building your Ubuntu Linux inVirtual Box
*Exact steps might have changed since publication of this lab

1. Open Virtual Box click the 'New' button (loos like a blue star-burst) and select the ISO file you downloaded and click open.
2. Fill out the initial question screen:
    a. Name: The name of you want to give your Ubuntu server
    b. Folder: The location you want your server stored
    c. ISO Image: Select the Ubuntu ISO you just downloaded
    d. Click skip unattended Install.
    e. Click next.
3. Select Base Memory and CPU. For this exercise I am doing 4 CPU and 4096 MB of RAM. Click next.
4. I am going to create a new virtual hard disk of 30 GB. (need 20 GB free per Greenbone)
5. Click finish.
6. Double click the Ubuntu server on the main VirtualBox screen to start your VM.
7. On the first screen select 'Try or Install Ubuntu' to start setup process
8. Choose language and click next.
9. Choose any accessibility options and click next.
10. Choose keyboard layout and click next.

11. Choose use wired connection for internet and click next.
12. Click install Ubuntu and click next.
13. Select Interactive installation and click next.
14. On the app selection page, select default selection and click next.
15. On 'Install recommended proprietary software do not select anything and click next.
16. On 'How do you want to install Ubuntu, select 'Erase disk and install Ubuntu' and click next.
17. On the create your account, provide username and password. Then click next.
18. Select your timezone and click next.
19. Click Install.
20. Wait while Ubuntu is installed.
21. At the end of the install process, you will be prompted to reboot.
22. Log in.
23. Press next to all of the 'first time' screen.
24. Click the Ubuntu button on the bottom and add the terminal to your quick launch.
25. Now to update our system. Open up the terminal and type 'sudo apt **update' and provide your password. Then enter the command** 'sudo apt upgrade'. Press 'Y' when prompted.
26. Open firefox and confirm you can reach the internet.
27. Power down your VM

Ubuntu Linux with VirtualBox as the Hypervisor

Download VirtualBox:

4. Go to: https://www.virtualbox.org/ and click downloads
5. Select your distribution to start your download
6. So much simpler than VMWare

Install Virtual Box:

10. Go to download location and double click. Accept UAC pop up.
11. Please note that exact install options might change as app is updated. This guide was created 8/24/2024.
12. On the first generic install screen click next.
13. The next screen allows you to customize your Virtual Box install. I kept with the default and clicked next.
14. Click Yes to the network interruption message.
15. Click yes if you get any missing dependencies.
16. At the next customization screen, I kept the default.
17. Click Install on the 'Ready to Install Screen'
18. Click Finish

Make sure that you are connected to the internet.

Now to this site and to download the ISO file for Ubuntu in the location of your choice:
https://ubuntu.com/download/desktop

Building your Ubuntu Linux inVirtual Box
*Exact steps might have changed since publication of this lab

28. Open Virtual Box click the 'New' button (loos like a blue star-burst) and select the ISO file you downloaded and click open.
29. Fill out the initial question screen:
    a. Name: The name of you want to give your Ubuntu server
    b. Folder: The location you want your server stored
    c. ISO Image: Select the Ubuntu ISO you just downloaded
    d. Click skip unattended Install.
    e. Click next.
30. Select Base Memory and CPU. For this exercise I am doing 2 CPU and 4096 MB of RAM. Click next.
31. I am going to create a new virtual hard disk of 20 GB.
32. Click finish.
33. Double click the Ubuntu server on the main VirtualBox screen to start your VM.
34. On the first screen select 'Try or Install Ubuntu' to start setup process
35. Choose language and click next.
36. Choose any accessibility options and click next.
37. Choose keyboard layout and click next.
38. Choose use wired connection for internet and click next.
39. Click install Ubuntu and click next.
40. Select Interactive installation and click next.
41. On the app selection page, select default selection and click next.
42. On 'Install recommended proprietary software do not select anything and click next.
43. On 'How do you want to install Ubuntu, select 'Erase disk and install Ubuntu' and click next.
44. On the create your account, provide username and password. Then click next.
45. Select your timezone and click next.
46. Click Install.
47. Wait while Ubuntu is installed.
48. At the end of the install process, you will be prompted to reboot.
49. Log in.
50. Press next to all of the 'first time' screen.
51. Click the Ubuntu button on the bottom and add the terminal to your quick launch.

52. Now to update our system. Open up the terminal and type ‘sudo apt **update’ and provide your password. Then enter the command** ‘sudo apt upgrade’. Press ‘Y’ when prompted.
53. Open firefox and confirm you can reach the internet.
54. Power down your VM
55. Jump over to the VirtualBox UI an click on your Ubuntu VM and click setting.
    a. Go to Shared Folders and click plus
        i. Folder Path: Point tofolder on host OS that has you PCAP files.
        ii. FolderName: it is auto filled in
        iii. MountPoint: QCC
        iv. Select Auto Mount
        v. Click OK
    b. Click on System and uncheck Optical
    c. Click Ok
    d. Restart Ubuntu VM

By default Virtual Box will put your VM in vlan local to your system and use NAT (Network Address Translation) in order to talk to systems on the outside. This is the same way your local router allows your system to communicate with the outside world. When you VM is turned off, you can go into settings -> network -> Attached to and set it to bridge so that your VM is on the exact same wireless you laptop is on but that will not buy us much for this exercise so lets keep it selected to NAT. (If you decide to put you Ubuntu Linux box on the same VLAN as your computer for the love of Pete Rose change your ‘ubuntu’ password)